

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented	☐ Application oriented	Batch ID:	
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project: RoadSense AI: Vision-Based Multi-Hazard Risk Assessment

Modern Advanced Driver Assistance Systems (ADAS) commonly use RGB dashboard cameras to detect pedestrians, vehicles, cyclists, and road-surface defects such as potholes. However, most existing systems generate separate alerts for each detected object without analyzing the interaction between multiple hazards or suggesting an appropriate driving action. Although object detection techniques have achieved high accuracy for individual hazard classes, they lack an integrated framework that combines multi-hazard detection, tracking, risk assessment, and decision support. This limitation increases the driver's cognitive load, especially in complex traffic scenarios where multiple hazards occur simultaneously. Therefore, there is a need for a real-time intelligent system that not only detects hazards but also prioritizes them and recommends safe driving actions.

This project proposes an RGB Vision-Based Multi-Hazard Risk Assessment and Driving Decision Support System that performs both hazard detection and scene-level reasoning using a standard RGB dashboard camera. The proposed pipeline employs a YOLOv11 object detector, fine-tuned through transfer learning, to detect pedestrians, vehicles, cyclists, and potholes. Detected objects are tracked across consecutive frames using ByteTrack to maintain object identity and estimate relative motion. A relative distance estimation module based on bounding-box size and perspective geometry, together with a pothole severity estimator, extracts additional information required for intelligent risk analysis.

The extracted information is processed by a Risk Scoring Module, which combines relative distance, object type, estimated motion, and pothole severity into a unified risk score for every detected hazard. The hazards are then ranked by priority, and a rule-based Decision Support Engine recommends one of five driving actions: Brake Immediately, Slow Down, Change Lane, Maintain Lane, or Proceed. The complete framework is implemented using Python, PyTorch, OpenCV, NumPy, Pandas, the Ultralytics YOLOv11 framework, ByteTrack, and a Streamlit/Flask dashboard for real-time visualization. CUDA acceleration is optionally used to improve inference speed on compatible GPU hardware.

The detection model is trained and evaluated using publicly available datasets for pedestrians, vehicles, cyclists, and potholes, together with curated Indian road footage for validation under realistic traffic conditions. The proposed system is designed to operate on live and recorded RGB video captured during normal daytime conditions and does not include autonomous vehicle control, thermal imaging, or custom sensing hardware. The effectiveness of the proposed framework will be evaluated using Mean Average Precision (mAP), Precision, Recall, F1-Score, and real-time performance exceeding 15 FPS, while the quality of the decision-support module will be validated using predefined expert-designed driving scenarios.

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1. [Base paper] Rubin, J., "Pothole Detection and Assessment on Highways Using Enhanced YOLO Algorithm With Attention Mechanisms," Advances in Civil Engineering, vol. 2025, Article ID 7911336, 16 pages, 2025. <https://doi.org/10.1155/adce/7911336>
2. Jagtap, A. M., Pawar, S., Shinde, P., Ahetti, P., and Raut, A., "YOLOv9-Based Visual Detection of Road Hazards and Anomalies: Potholes, Sewer Covers, and Manholes," International Journal for Research in Applied Science & Engineering Technology (IJRASET), 2025. <https://doi.org/10.22214/ijraset.2025.66913>
3. Lv, et al., "Application of YOLO Algorithm for Intelligent Transportation Systems: A Survey and New Perspectives," International Journal of Distributed Sensor Networks, 2025. <https://doi.org/10.1155/dsn/2859040>
4. Ultralytics, "Road safety with Ultralytics YOLO11: AI detection for safer streets," Ultralytics Blog, 2025. <https://www.ultralytics.com/blog/road-safety-with-ultralytics-yolo11-ai-detection-for-safer-streets>

Project Team Members

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